

EXPLAINABLE & TRUSTWORTHY AI · PROJECT · POLITECNICO DI TORINO

Concept discovery in medical VLMs: a rigorous characterization.

A faithful re-implementation of the MedConcept pipeline on BiomedCLIP —
IU X-Ray (~7k) and ROCOV2 (~80k, 10×). We characterise why SAE concept
discovery is non-identifiable, and what a deterministic alternative can do.

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THE PROBLEM

Medical VLMs are accurate but opaque.

Strong performance on pathology classification, segmentation and report generation — but the internal representations are high-dimensional, polysemic and unreadable. A serious concern in safety-critical clinical use, where classic post-hoc tools (saliency, attention) rarely resolve into semantic structure.

Ref: `main_extended.tex` §1 Introduction

APPROACH

Concept-based XAI: from supervised to unsupervised discovery.

Supervised

Concept Bottleneck Models (Koh 2020) and TCAV (Kim 2018) need a pre-defined, labelled concept set — exactly the limitation unsupervised discovery targets.

Impl: `src/autoencoder/`, `src/vocabulary_building/`, `src/evaluate_llm_judge.py`

Sparse Autoencoders

Bricken et al. (2023): dictionary-learning on activations yields monosemantic latents. The Top-K SAE (Gao 2024) sets L_0 exactly via top-k gating. We adopt it.

MedConcept (Haque 2026)

SAE extraction → alignment to a radiology vocabulary → an LLM-judge with Aligned/Unaligned/Uncertain metrics. The template we re-implement and stress-test.

Koh et al. 2020 — Concept Bottleneck Models Kim et al. 2018 — TCAV Bricken et al. 2023 — Towards Monosemanticity with Dictionary Learning Gao et al. 2024 — Scaling and Evaluating Sparse Autoencoders Haque et al. 2026 — MedConcept

Blankemeier et al. 2024 — Merlin 3D CT Foundation Model

RESEARCH GAPS · IN THE PAPER'S ORDER

Six gaps this work addresses.

GAP 1

Instability / non-identifiability

Sparse factorisations are not reproducible across seeds; no prior medical-VLM work reports seed-stability against a null.

GAP 4

Small-data robustness

SAEs need 10^5 – 10^6 activations; medical corpora are 2–3 orders smaller → dead features.

Ref: `main_extended.tex` §3 Research gaps

GAP 2

Clinical validity

A cosine-assigned name is cosmetic unless it tracks real pathology.

GAP 5

Flat concepts

Explanations are top-k lists that ignore anatomical / hierarchical structure.

GAP 3

Representation-location mismatch

SAE literature uses raw hidden states; reference pipelines fit on the already-projected, gap-bearing CLIP space.

GAP 6

Non-reproducible evaluation

Claims rest on single metrics whose chance floor is never reported; slot-wise overlap is degenerate under per-seed permutation.

BACKBONE & VOCABULARY

BiomedCLIP + paired lexicons.

Our backbone is BiomedCLIP. Its image encoder exposes a **768-d hidden state** that a frozen projection compresses into the **512-d** contrastive space we actually decompose. We name features against **domain-matched** vocabularies: **RadLex** for IU X-Ray (chest-specific), **MeSH** for ROCv2 (multimodal). And we correct the **modality gap** — the systematic offset between the image and text cones — before any naming or decomposition.

Impl: `src/embedding_extraction/extract_embeddings.py`, `src/vocabulary_building/build_vocabulary.py`

DATA

Two scales: IU X-Ray and ROCov2 (10×).

IU X-RAY · 1× · PRIMARY

The small-scale setting.

Split **by radiographic study**, so no study appears in both train and test — verified **zero** overlap, recomputed deterministically every run.

Impl: `src/utils.py` (`split_embeddings`) · anti-leakage test: `tests/unit/test_split.py`

ROCOV2 · 10× · REPLICATION

A ten-times
larger corpus.

Split by image. It exists to test one hypothesis head-on: **does more data cure non-identifiability?**

METHOD · THREE PATHS

Three decompositions, one SAE config.

01 · BASELINE

Top-K SAE 512-d

Faithful MedConcept re-implementation on the projected space.
Trained across **multiple seeds** — that is what makes a stability
metric possible. (Gap 1)

Impl: `src/autoencoder/sae_module.py` (SAEManager), `src/sae_hidden/`, `src/concept_discovery/spliece.py`

02 · PATH A

Top-K SAE 768-d

CLS hidden state *before* projection; bridge with frozen W_proj.
Tests the representation site. (Gap 3)

03 · PATH B

SPLiCE

Decomposition over a fixed dictionary matched to each corpus
(RadLex for IU X-Ray, MeSH for ROCov2) — no learning, so
deterministic. (Gap 1+4)

EVALUATION · GAP 6

A matched metric, against a subspace null.

Stability must be **permutation-invariant**: for each decoder row of one seed, take the best cosine match across the other seed. The null is conditioned on the decoder's **effective rank** (~357, not the ambient 512) — that controls for how concentrated the data manifold is. The old slot-wise Jaccard is not permutation-invariant, so it is degenerate (proved next).

Impl: `src/autoencoder/stability_analysis.py`, `src/sae_hidden/naming_hidden.py` (compute_random_baseline)

THE HEADLINE RESULT · GAP 1

1.81×

1.67× on IUX-Ray · 1.81× on ROCov2

The baseline SAE reconstructs its inputs well — but cross-seed **feature identity is absent**: best-match barely above the conditioned null, almost nothing matches strongly, naming stays low. Genuine non-identifiability, not a training bug.

RELABELING CONTROL · GAP 6

The slot-wise metric is floor-by-construction.

Rename one SAE's features with a random permutation — same network, same reconstruction — and the slot-wise Jaccard collapses to the **0.0077** floor, indistinguishable from two genuinely different SAEs. The matched metric correctly sees identity. Slot-wise is permutation noise, not a stability measure.

Impl: `scripts/relabeling_control.py - relabel_state_dict(sd, perm)`

 Cross-seed Jaccard matrix — every seed pair near the ~0.0077 floor

Cross-seed Jaccard matrix (IU X-Ray baseline). Mean ≈ 0.0077 — the analytical floor.

PATH A (768-D) VS BASELINE (512-D) · GAP 3

Moving the representation site is not enough.

Path A trains the SAE on the hidden state **before** the projection. It is genuinely healthier — far fewer dead features, better naming — but the permutation-invariant verdict does **not** change: still almost no features match strongly. This is **weak universality**: a shared subspace, not identical features. It still operates on the pooled CLS token, not patch tokens.

Impl: `src/sae_hidden/`, `scripts/run_path_a.py`

 Naming score distribution — most features' top-1 cosine below 0.5

Naming scores — top-1 feature↔vocab cosine. Most features sit well below 0.5: a cosine-assigned name is cosmetic.

THE SCALE HYPOTHESIS · GAP 4

10_x

ROCOv2 ~80k vs IU X-Ray ~7k

Ten times more data improves **training health** but **not identifiability** — every stability indicator stays flat. The binding constraint is the **representation site**, not data volume.

PATH B · SPLICE · GAP 1+4

Deterministic. Judge-ready.

SPLiCE decomposes each embedding into a sparse, non-negative combination over a **fixed domain-matched dictionary** (RadLex for IU X-Ray, MeSH for ROCOV2). The dictionary is never estimated, so the result is **deterministic by construction** — zero cross-seed instability. It scales to both corpora, and correcting the modality gap is the decisive fix, not the solver.

- **Fixed dictionary** — no estimation, no seeds, no instability.
- **Same output schema as the SAE** — the same judge can score it.
- **Honest caveat** — frequent terms are "mixed", a property of the supplied dictionary.

Impl: `src/concept_discovery/spliece.py` — `decompose_image()`

Dense, but weakly separated.

We cluster the active vocabulary into **concept families** with ontology ancestors (RadLex for IU X-Ray, MeSH for ROCOv2) and two degeneracy guards — leaf-root rejection, and a subtree canopy that rejects overly generic ancestors (e.g. "anatomical entity"). SPLiCE, the densest, cuts concepts-per-image **~1.75×** — but silhouette is weak (~ 0.02): the noisy vocabulary makes every cluster inherit a noisy ancestor.

Impl: `src/concept_discovery/organize.py` – `cluster_concepts()`, `annotate_radlex()`

A small upper tail is genuinely faithful.

Do **any** features track real pathology? Correlating activations against the ground-truth labels (with a shuffle null), a **small upper tail** is genuinely faithful — the strongest looks like a *mass*, clinically plausible. But those same features stay **unstable across seeds**: partially faithful, entirely non-reproducible.

Impl: `scripts/run_faithfulness.py` · `main_extended.tex` §5.6

LLM-AS-JUDGE · LOCAL PLAUSIBILITY

81.6%

SPLiCE MedGemma · 79.6% null-k

SPLiCE scores **high local plausibility** under a medical-tuned judge — but **almost nothing** under a general one. **Random-k null explanations score comparably** (79.6%), indicating weak discriminability. The Aligned rate is a property of the **judging model** as much as of the explanation. And this local plausibility is **orthogonal** to global identifiability.

CONCLUSIONS

Three lessons that generalise.

Null + permutation-invariant

Interpretability claims about discovered concepts must be calibrated against analytical nulls **and** a permutation-invariant statistic. Slot-wise overlap is degenerate.

Scale is the last suspect

For a factorisation with limited identifiability, data scale is the least likely culprit: 10× changed training health, not identifiability.

Partially-negative evaluation

An honest, partially-negative evaluation is more useful than a curated one. The defect is localised (representation site), not global.

Ref: `main_extended.tex` §6 – 4 contributions: relabeling control · matched metric + null · scale refutation · partially-negative eval

LIMITATIONS & FUTURE WORK

SAE on the patch-token residual stream.

The diagnosis is predictive: decomposing BiomedCLIP **patch-token** residual stream (a mid layer, not the pooled CLS) should restore identifiability — the experiment our characterisation points to as the next step.

- **Limitation (i):** LLM-judge is model-dependent; random-k nulls score comparably.
- **Limitation (ii):** predicted mitigation is SAE on patch-token residual stream.
- **Limitation (iii):** ROCov2 confounds scale with domain; PadChest (~160k chest X-rays) is the clean test.

Thank you · questions — Marc'Antonio Lopez · Nicolò Colle · Carmine Francesco Benvenuto